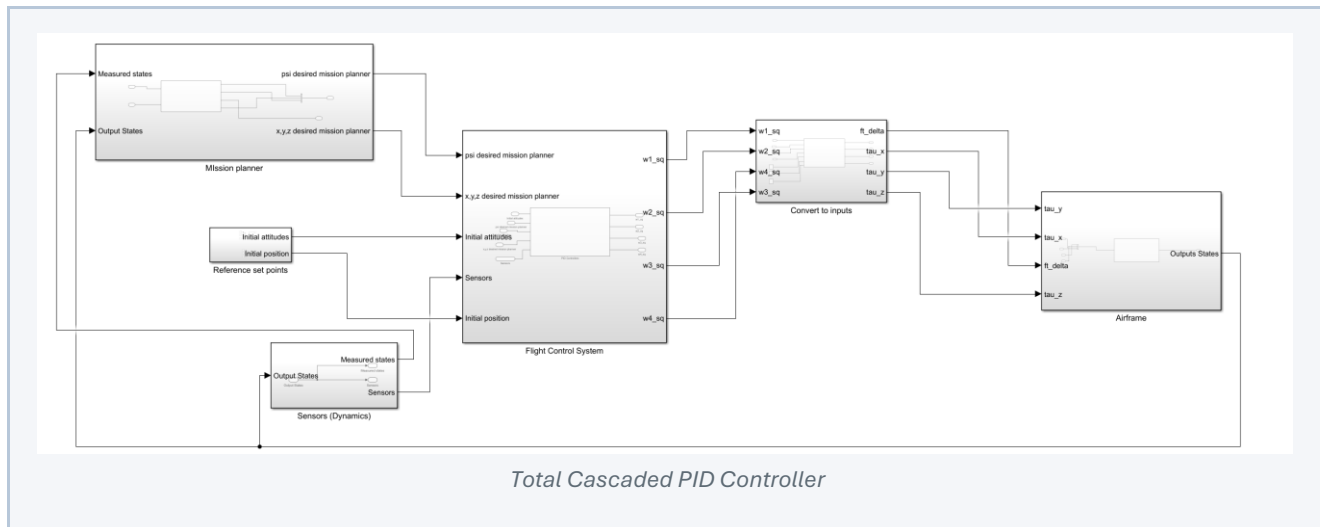


FLIGHT CONTROL PROJECT PORTFOLIO

Cascaded PID Controller Design in Simulink

Multicopter Attitude and Angular-Rate Control



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Controller Design • Simulation • Validation

Project Overview

This project developed a cascaded PID attitude controller for a Quadcopter and implemented the control architecture in Simulink. The design separates the problem into a fast inner loop that stabilizes angular rate and a slower outer loop that tracks commanded attitude. This structure reflects the physical hierarchy of a flight-control system and provides a clear path from rigid-body dynamics to a simulation-ready controller.

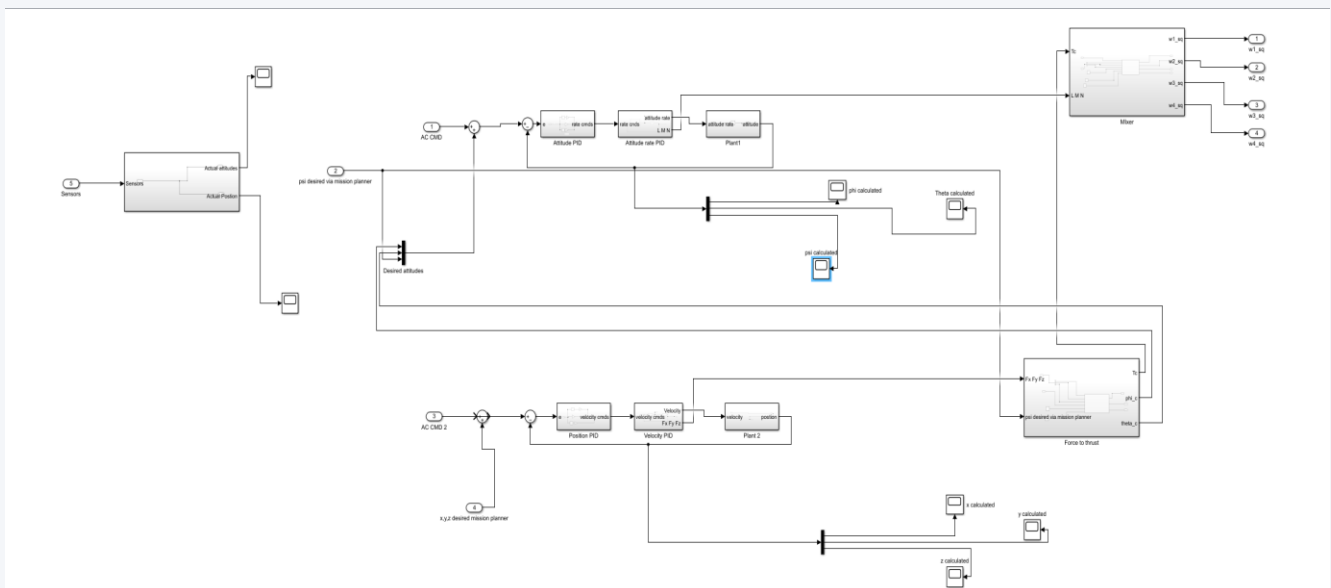
Engineering Objective

Design a controller that commands propellor torques to reach a desired position and orientation while maintaining stable, well-damped angular-rate behavior. The controller was designed by matching the closed-loop dynamics to selected second-order targets and then translating the resulting gains into a cascaded Simulink implementation.

INNER LOOP	OUTER LOOP	DESIGN METHOD
Angular rate PID	Attitude PID	Coefficient matching

Control Architecture

The outer loop compares the commanded position and orientations with the current states. These inner loops produce the necessary body forces and torques to achieve flight goals. These values enter the mixer which calculates how much force the 4 propellers need to achieve. This is essential to ensure the desired values are correctly channeled into the right control surfaces.



Outer PID Loops

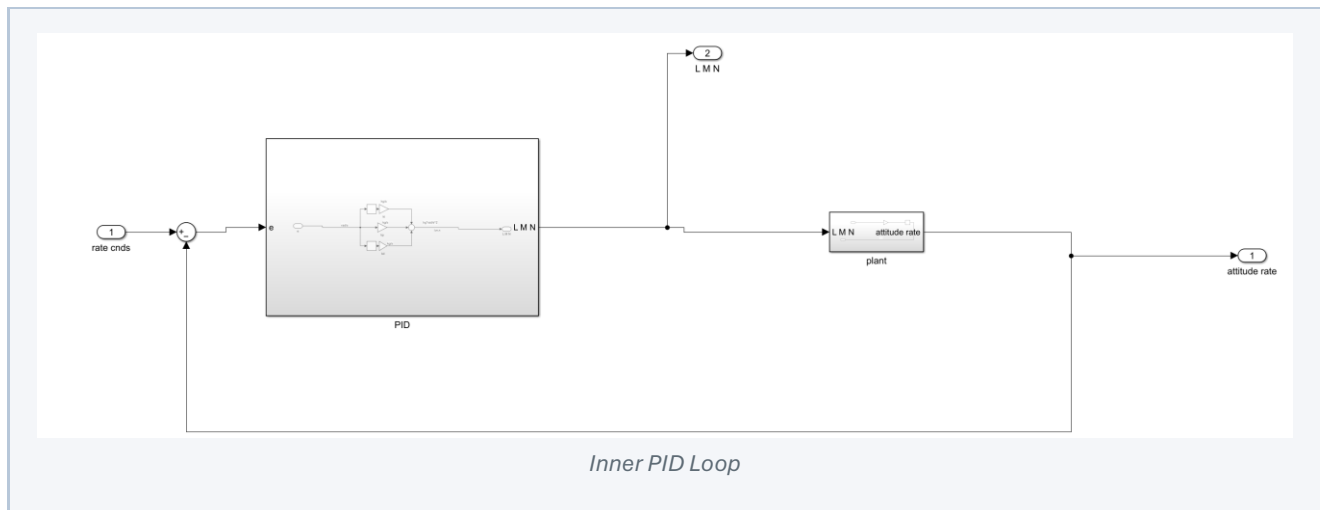
Modeling and Design Basis

Roll-Axis Dynamics

For the roll axis, the rotational equation of motion is $\dot{p} = L / I_x$. In the Laplace domain, the plant from roll torque to roll rate is $p(s)/L(s) = 1/(I_x s)$. The roll inertia used for the design was $I_x = 5.897 \times 10^{-3} \text{ kg}\cdot\text{m}^2$. This is extended to the other attitude rates and velocity loops.

Inner-Loop Rate Controller

The inner controller is a full PID law: $C_r(s) = K_{dr} s + K_{pr} + K_{ir}/s$. Closing the loop around the rate plant produces a second-order characteristic equation, $(I_x + K_{dr})s^2 + K_{pr} s + K_{ir} = 0$. This form makes the gains directly relatable to a desired damping ratio and natural frequency.



Performance Targets and Gain Selection

The rate loop was assigned $\zeta_r = 0.8$ and $\omega_r = 15 \text{ rad/s}$ to make it substantially faster than the attitude loop. Derivative gain was selected first as $K_{dr} = 0.25 I_x$. The remaining gains were obtained by matching the second-order coefficients: $K_{pr} = (I_x + K_{dr}) \cdot 2\zeta_r \omega_r$ and $K_{ir} = (I_x + K_{dr}) \cdot \omega_r^2$.

ζ_r	ω_r	K_{dr}	K_{pr} / K_{ir}
0.8	15 rad/s	1.474×10^{-3}	0.177 / 1.659

Outer-Loop Attitude Design

The inner-loop rate dynamics were combined with the kinematic relation $\dot{\phi} = p$ and an outer PID controller. This produces a fourth-order closed-loop system. Because only three outer-loop gains are available, all four polynomial coefficients cannot be independently matched. The design therefore uses approximate pole placement by matching three key coefficients of the desired fourth-order polynomial.

Time-Scale Separation

The outer loop was intentionally made slower, with $\zeta\phi = 0.8$ and $\omega\phi = 3$ rad/s. The desired overall dynamics were constructed from the product of the inner and outer target polynomials: $(s^2 + 24s + 225)(s^2 + 4.8s + 9)$. This preserves the important physical intent: rapid rate stabilization and smoother attitude tracking.

Final Attitude Gains

Kd ϕ 0.2083	Kp ϕ 5.00	Ki ϕ 9.375
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Design Tradeoff

The achieved normalized characteristic polynomial matches the s^3 , s^1 , and constant terms of the desired polynomial exactly. The s^2 term is slightly different, which is expected because the fourth-order system has only three independently adjustable outer-loop gains. This is an intentional engineering tradeoff rather than an error: the design prioritizes stable separated time scales over exact placement of every pole.

$$\frac{\phi(s)}{\phi_c(s)} = \frac{K_{d\phi}K_{dr}s^4 + (K_{d\phi}K_{pr} + K_{p\phi}K_{dr})s^3 + (K_{d\phi}K_{ir} + K_{p\phi}K_{pr} + K_{i\phi}K_{dr})s^2 + (K_{p\phi}K_{ir} + K_{i\phi}K_{pr})s + K_{i\phi}K_{ir}}{[(I_x + K_{dr}) + K_{d\phi}K_{dr}]s^4 + [K_{pr} + K_{d\phi}K_{pr} + K_{p\phi}K_{dr}]s^3 + [K_{ir} + K_{d\phi}K_{ir} + K_{p\phi}K_{pr} + K_{i\phi}K_{dr}]s^2 + [K_{p\phi}K_{ir} + K_{i\phi}K_{pr}]s + K_{i\phi}K_{ir}}$$

Above is the derived pole placement equation

Validation and Results

This section is designed for the evidence generated after running the Simulink model. Use the same command input and axis scales when comparing controller revisions so the effects of tuning remain easy to interpret. We will ensure our desired x,y,z location and angles are all converging quickly and accurately. We desire the following:

$X = 5$

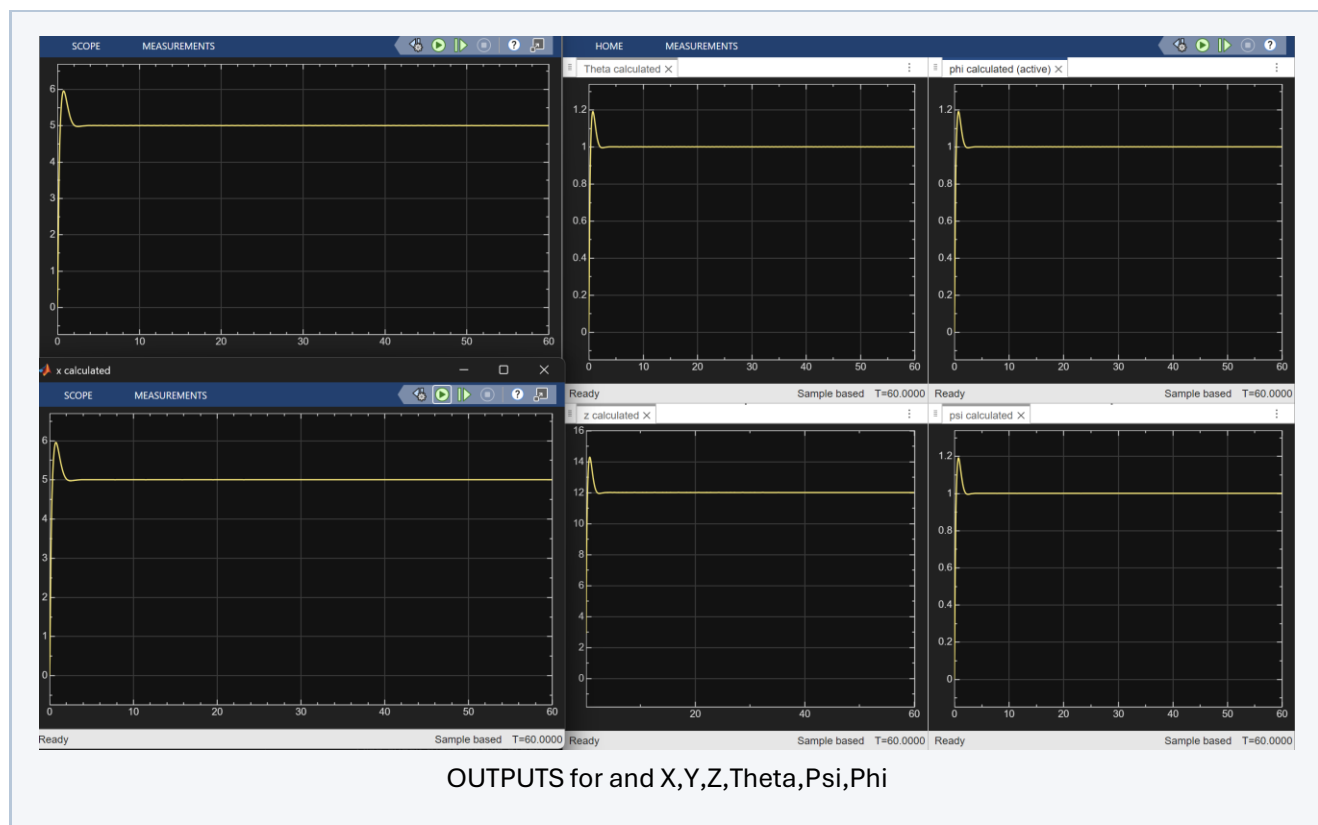
$Y = 5$

$Z = 12$

Roll = 1 degree

Pitch = 1 degree

Yaw = 1 degree



As we see above, we have quickly converged to our desired value with a quick rise time and relatively low overshoot. This shows the cascaded PID controller is working as planned. It also ensures the pole placements decisions were accurate and have small error from the approximations made.